1.

[*] HW4_1.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{

    int num[100];
    int number,i;

    cout << "Enter a number between 1 & 100:";
    cin >> number;
    cout << "Factors are:" << endl;

    for(i=1;i<=number;++i)
    {

        if(number%i==0)
        {
            num[i]=i;
            cout << num[i] << " ";
        }
    }

    return 0;
}
```

```
Enter a number between 1 & 100:78
Factors are:
1 2 3 6 13 26 39 78
--------------------------------
Process exited after 3.883 seconds with return value 0
請按任意鍵繼續 . . .
```

2.

[*] HW4_2.cpp

```cpp
#include <iostream>
#include<utility>
using namespace std;

int main()
{
    int num1,num2,m,n;

    cout << "Enter the first number (positive):";
    cin >> m;
    num1 = m;
    cout << "Enter the second number (positive):";
    cin >> n;
    num2 = n;

    while(n!=0)
    {
        m = m % n;
        if(m < n)
            swap(m,n);
    }

    cout << "The greatest common divisor (GCD) between " << num2 << " and " << num1 << " is:" << m;

    return 0;
}
```

```
Enter the first number (positive):9
Enter the second number (positive):12
The greatest common divisor (GCD) between 12 and 9 is:3
--------------------------------
Process exited after 7.941 seconds with return value 0
請按任意鍵繼續 . . .
```

```
Enter the first number (positive):7
Enter the second number (positive):11
The greatest common divisor (GCD) between 11 and 7 is:1
--------------------------------
Process exited after 6.296 seconds with return value 0
請按任意鍵繼續 . . .
```

```
Enter the first number (positive):140
Enter the second number (positive):12
The greatest common divisor (GCD) between 12 and 140 is:4
--------------------------------
Process exited after 5.962 seconds with return value 0
請按任意鍵繼續 . . .
```

3.

HW4_3.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{

    int num[10];
    int number,i,s;

    cout << "Enter a positive number:";
    cin >> number;

    for(i=0;number>0;++i)
    {
        num[i]=number%10;
        number/=10;
    }

    cout << "The reversed number is:";

    for(s=0;s<i;++s)
        cout << num[s];

    return 0;
}
```

```
Enter a positive number:145
The reversed number is:541
--------------------------------
Process exited after 3.365 seconds with return value 0
請按任意鍵繼續 . . .
```

```
Enter a positive number:9
The reversed number is:9
--------------------------------
Process exited after 7.601 seconds with return value 0
請按任意鍵繼續 . . .
```

```
Enter a positive number:14567
The reversed number is:76541
--------------------------------
Process exited after 3.453 seconds with return value 0
請按任意鍵繼續 . . .
```

4.

[*] HW4_4.cpp

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    float table[2][100];
    int p=0;


    for( int i=0 ; i<100 ; ++i )
    {
        table[0][i] = i;
        table[1][i] = i * 180.0 / 100.0 + 32;
    }


    for( int i=0 ; i<5 ; ++i )
        cout << setw(4) << "Ce" << setw(8) << "Fahr";
    cout << endl;

    for( int i=0; i<5 ; ++i )
        cout << setw(4) << "--" << setw(8) << "-----";
    cout << endl;
```

```cpp


    for( int j=0 ; j<20 ; ++j )
    {
        p=j;
        for( int i=0 ; i<5 ; ++i )
        {
            cout << fixed << setprecision(0) << setw(4) << table[0][p];
            cout << fixed << setprecision(2) << setw(8) << table[1][p];
            p += 20;
        }
        cout << endl;
    }

    return 0;
}
```

```
 Ce    Fahr  Ce    Fahr  Ce    Fahr  Ce    Fahr  Ce    Fahr
 --    -----  --    -----  --    -----  --    -----  --    -----
  0   32.00  20   68.00  40  104.00  60  140.00  80  176.00
  1   33.80  21   69.80  41  105.80  61  141.80  81  177.80
  2   35.60  22   71.60  42  107.60  62  143.60  82  179.60
  3   37.40  23   73.40  43  109.40  63  145.40  83  181.40
  4   39.20  24   75.20  44  111.20  64  147.20  84  183.20
  5   41.00  25   77.00  45  113.00  65  149.00  85  185.00
  6   42.80  26   78.80  46  114.80  66  150.80  86  186.80
  7   44.60  27   80.60  47  116.60  67  152.60  87  188.60
  8   46.40  28   82.40  48  118.40  68  154.40  88  190.40
  9   48.20  29   84.20  49  120.20  69  156.20  89  192.20
 10   50.00  30   86.00  50  122.00  70  158.00  90  194.00
 11   51.80  31   87.80  51  123.80  71  159.80  91  195.80
 12   53.60  32   89.60  52  125.60  72  161.60  92  197.60
 13   55.40  33   91.40  53  127.40  73  163.40  93  199.40
 14   57.20  34   93.20  54  129.20  74  165.20  94  201.20
 15   59.00  35   95.00  55  131.00  75  167.00  95  203.00
 16   60.80  36   96.80  56  132.80  76  168.80  96  204.80
 17   62.60  37   98.60  57  134.60  77  170.60  97  206.60
 18   64.40  38  100.40  58  136.40  78  172.40  98  208.40
 19   66.20  39  102.20  59  138.20  79  174.20  99  210.20

--------------------------------
Process exited after 0.2952 seconds with return value 0
請按任意鍵繼續 . . .
```